UI/UX Design Document: GREENIE Autonomous Fleet Management System

1. Design Vision & Concept

The objective is to evolve the existing GREENIE dashboard from a static tracking portal into a dynamic, autonomous-ready command center. By fusing the structural layout of your current system (Images 3–7) with the advanced telemetry of autonomous vehicle interfaces (like Fleeet.so), the UI will allow dispatchers to monitor AI-driven routes, view real-time vehicle camera feeds, and track electric/autonomous vehicle battery levels seamlessly.

2. Visual Language & Design System

Based on your uploaded UI screenshots, the system will retain its clean, professional aesthetic while introducing high-tech elements suited for AI operations.

- **Color Palette:**
    - **Primary Base:** Deep Navy Blue (#1E293B) for the left sidebar and top navigation to maintain brand consistency.
    - **Background:** Off-white/Light Gray (#F8FAFC) for the main workspace to ensure high contrast for data cards.
    - **Accents:** Gold/Yellow (#F59E0B) for active tab states and primary action buttons.
    - **Status Indicators (Semantic):** Emerald Green (Active/Charging), Amber (Idle/Low Battery), Rose Red (Maintenance/AI Alert).
    - **Theme Toggle:** Introduce a **Dark Mode** (dark slate backgrounds with neon accents), which is an industry standard for autonomous monitoring and reduces eye strain in control rooms.
- **Typography & Layout:**
    - Clean, sans-serif fonts (like Inter or Roboto) for high legibility of data grids.
    - **Modular Card System:** Keep the existing rounded-corner white cards with subtle drop shadows (as seen in Image 4 and 6) to house individual widgets.

3. Core Interface Components & Upgrades

A. Operations Dashboard (Upgrading Image 4)

Current State: Displays Total Units, Active Units, Pending Collections, and Fleet Status Distribution.

- **Fleeet.so Integration:**
    - Convert "Total Units" to differentiate between *Human-Driven* and *Autonomous Bots*.
    - **New Widget - AI Fleet Efficiency:** Add a card displaying "AI Fuel/Energy Saved Today" and "Route Optimization %".
    - **Charging Status Ring:** Replace the simple "Fleet Status Distribution" donut chart with a concentric ring chart showing real-time fleet battery/charging levels (e.g., >80% charged, charging, critically low).

B. Live Monitoring & Telemetry Map (Upgrading Image 5)

Current State: A static OpenStreetMap view with colored status pins (Active, Idle, Maintenance).

- **Autonomous Tracking:**
    - **Dynamic Routing Lines:** Display the AI-generated CVRP (Capacitated Vehicle Routing Problem) paths as glowing lines on the map.
    - **Live Camera Feed Module (PiP):** Inspired by your Python object-detection script (Image 2), clicking on any active autonomous truck pin will open a Picture-in-Picture (PiP) window. This window will stream the vehicle's dashcam or cargo-bay camera, overlaying real-time AI bounding boxes (e.g., showing *concrete 89%*, *metal 47%* as the waste is loaded).
    - **Telemetry Sidebar:** When a vehicle is selected, the right-hand panel shifts from "Site Status" to "Vehicle Telemetry," showing live speed, battery degradation, payload weight (from load cells), and autopilot status.

C. Fleet & Autonomous Vehicle Registry (Upgrading Image 7)

Current State: Standard table with ID, Plate No., Model, Driver, Status, Mileage, and Last Service.

- **Autonomous Metrics:**
    - **Driver Column Update:** Change the "Driver" column to "Operator / Mode" (e.g., *Autonomous*, *Remote Override*, or *Human Assisted*).
    - **New Columns:** Add "Battery Health (%)" and "AI Sensor Status" (showing a green checkmark if LiDAR and cameras are fully operational).
    - **Predictive Maintenance Alerts:** Highlight specific rows in light red if the ML model predicts an imminent component failure, replacing static "Last Service" dates with "Predicted Service Required (Days)."

D. AI Segregation & Transfer Station Analytics (Upgrading Image 6)

Current State: Shows Total Waste Collected, Avg Per Site, Revenue, and Waste by Type bar charts.

- **Integration with Image 2 (Object Detection):**
    - **Live Segregation Feed:** Create a dedicated tab within Analytics that directly ports the video feed from the transfer station's conveyor belt.
    - **Automated Auditing:** Link the AI bounding box data (Image 2) directly to the "Waste by Type" bar chart (Image 6). As the camera detects *hard plastic (81%)* or *timber (64%)*, the analytics dashboard updates the tonnage and projected revenue in real-time without human data entry.

4. Interaction Design & User Flow

- **Remote Override Controls:** For autonomous units, add a "Take Control" button on the vehicle detail page. This allows a remote dispatcher to safely halt or reroute the vehicle in case of a blocked road or emergency.

- **Real-Time Data WebSockets:** The UI must feel "alive." Map pins should glide smoothly rather than jumping, and metrics like "Revenue Generated" and "Waste Collected" should tick upward continuously as the AI sorts materials at the station.
- **Push Notifications:** Slide-in toast notifications at the top right of the screen for critical AI alerts (e.g., *"Unit V-003 rerouted due to heavy traffic"* or *"High moisture content detected at Coimbatore North Depot"*).